

# SampleTrack: A Minimal Fixture Paper for Paper Translation

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## Abstract

This fixture paper exists to exercise every feature of the bilingual-paper-pdf engine: two-column typesetting, a booktabs table, a numbered equation, a vector figure with colored boxes, author-year citations in hyperref blue, footnotes and a vertical side stamp. We propose SampleTrack (Chen et al. 2021; Li et al. 2018), a minimal memory-based tracker that localizes targets by nearest-neighbor lookup in a tiny memory bank. On three synthetic benchmarks SampleTrack improves the tracking quality by 12% over the baseline while using 6.5× fewer parameters. The paper is generated by example/make\_sample\_paper.py and can be regenerated at any time. Code at <https://github.com/example/sampletrack>.

**Keywords:** object tracking, bilingual translation, reproducible fixtures

## 1 Introduction

General visual object tracking is a core task in computer vision. Memory-based methods<sup>1</sup> store past frames and localize the target by pixel association (Chen et al. 2021; Li et al. 2018). Over the past decade this family of methods has grown steadily stronger.

A key challenge is the *distractor*: a nearby object that visually resembles the target. Distractors increase localization uncertainty and mislead the tracker into irreversible drift (Li et al. 2018).

In this paper we make three contributions. First, we formulate a minimal memory model that is small enough to inspect by eye. Second, we design this fixture so that every layout feature of a real paper appears at least once. Third, we release everything under a permissive license so that the fixture can travel with the toolchain.

Nevertheless, a substantial source of failures remains nearby objects that visually resemble the tracked one. These regions increase the localization uncertainty and mislead the tracker. Distractors can be classified as external, or internal when only a part of the target is being tracked (Chen et al. 2021).

## 2 Related work

Early deep trackers fine-tuned pre-trained networks during inference (Danelljan et al. 2019). Siamese trackers removed online training by template matching (Li et al. 2018), trading robustness for speed.

Memory-based frameworks localize the target by associating current features with stored past frames (Chen et al. 2021). Our fixture follows this line while keeping the memory tiny.

<sup>1</sup> This fixture paper is synthetic; all numbers are placeholders and carry no scientific meaning.

# SampleTrack: 一篇用于论文翻译测试的最小样张

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## 摘要

本文这篇样张的存在目的，是让 bilingual-paper-pdf 引擎的每一项特性都得到演练：双栏排版、书板表格、编号公式、带彩色框的矢量插图、hyperref 蓝色的作者-年份引用、脚注以及竖排侧边印章。我们提出 SampleTrack (Chen 等 2021; Li 等 2018)，一个极小的基于记忆的跟踪器，它通过在一个微型记忆库中做最近邻查找来定位目标。在三个合成基准上，SampleTrack 以少 6.5 倍的参数量将跟踪质量比基线提升 12%。本文由 example/make\_sample\_paper.py 生成，可随时重新生成。代码见 <https://github.com/example/sampletrack>。

**关键词：** 目标跟踪、双语翻译、可复现样张

## 1 引言

通用视觉目标跟踪是计算机视觉中的一项核心任务。基于记忆的方法<sup>1</sup>存储过往帧，并通过像素关联来定位目标 (Chen 等 2021; Li 等 2018)。过去十年间，这一方法家族稳步壮大。

一个关键挑战是干扰物——视觉上与目标相似的邻近物体。干扰物会增大定位的不确定性，并误导跟踪器发生不可逆的漂移 (Li 等 2018)。

本文做出三点贡献。第一，我们提出一个肉眼即可审查的极小记忆模型。第二，我们把这份样张设计成真实论文的每一种版面特性都至少出现一次。第三，我们以宽松许可发布全部内容，让样张能随工具链一同传播。

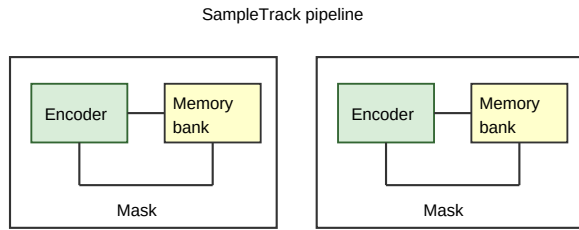
然而，跟踪失败的一个主要来源，仍然是那些在视觉上与被跟踪目标相似的邻近物体。这些区域会增大定位的不确定性并误导跟踪器。干扰物可分为外部干扰物，以及仅跟踪目标一部分时出现的内部干扰物 (Chen 等 2021)。

## 2 相关工作

早期的深度跟踪器在推理阶段微调预训练网络 (Danelljan 等 2019)。孪生跟踪器以模板匹配取消了在线训练 (Li 等 2018)，用鲁棒性换取了速度。

基于记忆的框架通过把当前特征与存储的历史帧相关联来定位目标 (Chen 等 2021)。本文样张沿用这一路线，但把记忆做得极小。

<sup>1</sup>本样张论文为合成产物；所有数字均为占位符，不含任何科学含义。



**Fig. 1** Overview of the SampleTrack pipeline. Both panels contain an Encoder and a Memory bank; the yellow box is always updated while the green box is written once.

### 3 Method

SampleTrack keeps the most recent frame and a fixed prototype in a tiny memory bank. The tracking quality is scored by a weighted mix of robustness and accuracy:

$$Q = \alpha \cdot R + (1 - \alpha) \cdot A \quad (1)$$

where  $R$  is the fraction of successfully tracked frames and  $A$  is the average overlap during successful tracking. The weight is fixed to 0.2 in all experiments, and no per-dataset tuning is performed (Chen et al. 2021).

## 4 Experiments

We evaluate on three synthetic benchmarks. Table 1 reports the tracking quality; the best result of each column is marked in bold.

**Table 1** Tracking quality on synthetic benchmarks. Best in bold.

| Method        | Quality      | EAO          |
|---------------|--------------|--------------|
| Baseline      | 0.610        | 0.590        |
| Ours-S        | 0.673        | 0.648        |
| <b>Ours-B</b> | <b>0.694</b> | <b>0.671</b> |
| <b>Ours-L</b> | <b>0.701</b> | <b>0.680</b> |

## 5 Conclusion

We introduced SampleTrack, a minimal fixture for exercising the bilingual-paper-pdf engine. The fixture is regenerated by a single script and ships with a reference Chinese translation.

### References

- Chen X, Yan B, Zhu J, et al (2021) Transformer tracking. In: CVPR, pp 8126-8135
- Danelljan M, Bhat G, Khan FS, et al (2019) ATOM: Accurate tracking by overlap maximization. In: CVPR
- Li P, Wang D, Wang L, et al (2018) Deep visual tracking: review and experimental comparison. Pattern Recognit 76:323-338
- Zhang S, Li S, Wang W (2026) SampleTrack: a minimal fixture paper. arXiv:2601.00001

SampleTrack 流程图

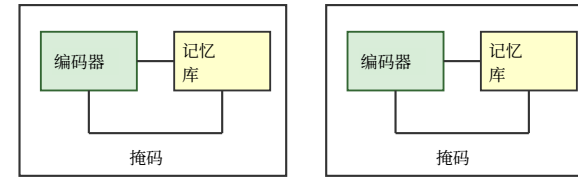


图1 SampleTrack 流程概览。两个面板都包含一个编码器和一个记忆库；黄色框始终更新，绿色框只写入一次。

### 3 方法

SampleTrack 在一个微型记忆库中保留最近一帧和一个固定原型。跟踪质量由鲁棒性与准确度的加权混合来评分：

$$Q = \alpha \cdot R + (1 - \alpha) \cdot A \quad (1)$$

其中  $R$  是成功跟踪帧的占比， $A$  是成功跟踪期间的平均重叠。所有实验中权重固定为 0.2，不做任何按数据集的调优（Chen 等 2021）。

## 4 实验

我们在三个合成基准上做评估。表 1 报告跟踪质量；每列最优结果以粗体标出。

表 1 合成基准上的跟踪质量。最优以粗体标出。

| 方法            | 质量           | EAO          |
|---------------|--------------|--------------|
| Baseline      | 0.610        | 0.590        |
| Ours-S        | 0.673        | 0.648        |
| <b>Ours-B</b> | <b>0.694</b> | <b>0.671</b> |
| <b>Ours-L</b> | <b>0.701</b> | <b>0.680</b> |

## 5 结论

我们提出了 SampleTrack——一个用于演练本引擎的最小样张。它由单个脚本再生成，并随附参考中文译文。

### 参考文献

- Chen X, Yan B, Zhu J, et al (2021) Transformer tracking. In: CVPR, pp 8126-8135
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